

Statistical Learning

Introduction to Machine Learning Foundations

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What is Statistical Learning?

Definition

Statistical learning refers to a set of tools used for understanding data and learning relationships between variables.

$$Y = f(X) + \epsilon$$

- X : input variables / predictors / features
- Y : response or output variable
- $f(X)$: systematic information learned from data
- ϵ : random error (noise)

Main Objective

- Predict future outcomes accurately
- Understand relationships hidden in data

Applications

- Recommendation systems
- Spam filtering
- Medical diagnosis

Why Estimate f ?

Goal 1: Prediction

- Use input variables X to predict output Y
- Focus on prediction accuracy
- Estimated function:

$$\hat{Y} = \hat{f}(X)$$

Examples:

- House price prediction
- Weather forecasting

Goal 2: Inference

- Understand relationship between X and Y
- Which predictors are important?
- How does Y change with X ?

How Do We Estimate f ?

Two major approaches are used to estimate f :

Parametric Methods

- Assume a fixed functional form
- Example:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- Simple and interpretable

Non-Parametric Methods

- No strict assumption on model form
- More flexible
- Can capture complex patterns

The Trade-Off Between Prediction Accuracy and Model Interpretability

Different statistical learning methods provide different levels of flexibility and interpretability.

Interpretable Models

- Simple models are easier to understand and explain. In linear regression, the effect of each predictor can be directly interpreted using model coefficients.
- These models are useful in domains where explanation is important, such as healthcare, finance, and scientific research.

Flexible Models

- Flexible methods can capture complex and nonlinear relationships in data. These methods often provide higher prediction accuracy.
- However, as flexibility increases, interpretability decreases. Examples include Deep Neural Networks and Ensemble Methods.

Higher Flexibility $\Rightarrow$ Higher Prediction Accuracy $\Rightarrow$ Lower Interpretability

Types of Learning

Supervised Learning

- Input + Output available
- Learn mapping from X to Y

Examples:

- Regression
- Classification

Unsupervised Learning

- Only input data available
- Find hidden structures/patterns

Regression vs Classification

Regression

- Predict continuous values
- Examples:
 - Salary
 - Temperature

Classification

- Predict categories/classes
- Examples:
 - Spam / Not Spam
 - Fraud / Not Fraud

Measuring quality of fit

After training a statistical learning model, we evaluate how well it performs on unseen data.

Regression Setting

Model performance is commonly measured using Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- Measures average squared prediction error
- Lower MSE indicates better predictions

Classification Setting

$$Error\ Rate = \frac{1}{n} \sum_{i=1}^n I(y_i \neq \hat{y}_i)$$

- Measures proportion of incorrect predictions
- Lower error rate implies better classification performance

Training Error vs Test Error

Training Error

- Error on seen/training data

Test Error

- Error on unseen data
- Real measure of model performance

Important:

Low Training Error $\neq$ Good Generalization

2.2.2 The Bias-Variance Trade-Off

$$E(y_0 - \hat{f}(x_0))^2 = \text{Var}(\hat{f}(x_0)) + [\text{Bias}(\hat{f}(x_0))]^2 + \text{Var}(\epsilon)$$

This equation shows that prediction error consists of three components:

Bias

- Bias refers to the error introduced by approximating a real-world problem using a simplified model.
- High bias models are too simple and may fail to capture important patterns in the data.
- Such models generally lead to **underfitting**.

Variance

- Variance measures how much the estimated model $\hat{f}$ would change if we used a different training dataset.
- High variance models are highly sensitive to small fluctuations in the training data. Such models generally lead to **overfitting**.

Understanding the Bias-Variance Trade-Off

- Increasing model flexibility usually decreases bias. However, increasing flexibility also increases variance.
- Therefore, there exists a trade-off between bias and variance.

Low Bias + High Variance

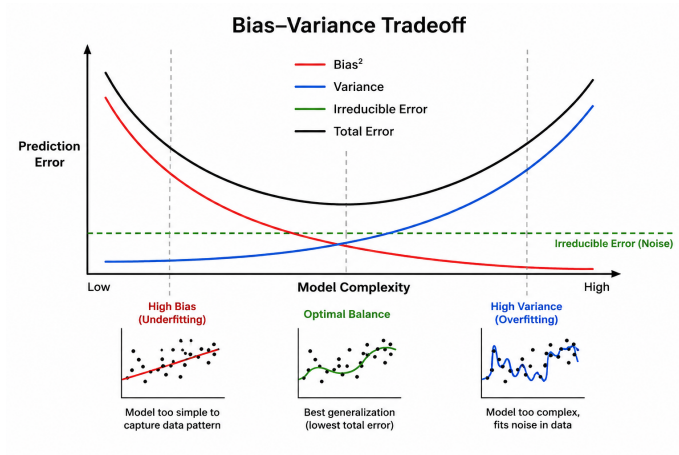
- Very flexible models. May overfit the training data

High Bias + Low Variance

- Very simple models. May underfit the data

Main Objective

Choose a model complexity that minimizes the **Test Mean Squared Error**.



Summary

- Statistical learning learns patterns from data
- Why and how do we estimate f ?
- Supervised vs unsupervised learning
- Regression vs classification
- Training vs test error
- Bias-variance tradeoff is fundamental

Questions?